

INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING 2019,
ICRTAC 2019

Classification of Neurodegenerative Disease Stages using Ensemble Machine Learning Classifiers

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Abstract

Existing research works for Alzheimer's disease (AD) can predict the prevalence of disease only after the advancement of the disease. With these existing prediction models, it is possible only to reduce and delay the symptoms of the disease. The exact usefulness is when the presence of the disease is identified at an early stage and this early detection makes a great impact in subjects' recovery. Thus, early detection of controls at high risk of development of Alzheimer's disease is of a key objective of the proposed work. Existing machine learning and deep learning algorithms derive only limited predictive accuracy. Also, they derive results based on expensive machine learning algorithms that had hard-to-collect features and classifying becomes complex with numerous overfitting in choosing decision boundaries. The proposed study intends to develop a learning algorithm for the prediction of Alzheimer's disease at an early stage. It also classifies the features if the subjects with Mild Cognitive impairment (MCI) and Pre-Mild Cognitive Impairment (Pre-MCI) has the likelihood to develop Alzheimer's disease. A dataset of AD controls was used to train different machine learning algorithms. Onset information like social behavior, demographic characteristics, neurological test scores, clinical cardiovascular index, and brain atrophy ratio can also be used as the extract predictor. A validation procedure was applied to identify a relevant subset of predictors. The conversion to AD in MCI and Pre MCI subjects are based only on non-invasively and effectively collectible predictors.

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Peer-review under responsibility of the scientific committee of the INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING 2019.

Keywords: Alzheimer's disease, Pre-Mild Cognitive impairment, Mild Cognitive impairment, Normal controls.

1. Introduction

Alzheimer's disease is a neurodegenerative disease that affects 47 million people worldwide and will affect around 150 million people by 2050. The financial resource will be spent on AD is assumed to be 800 billion dollars that are early 1% of the world's total GDP. Alzheimer's pathology, when occurring in later stages of old elderly individuals, the symptoms of which prevail before 20 years of the onset of the disease[1]. Hence detection at a very early stage has been the existing research problem. During the final onset of disease, brain neurons are accumulated with tau proteins called tangles and plaque proteins called beta-amyloid proteins. Before the beginning of the disease, the genetic characteristic includes brain atrophy and the increased level of tau and plaques in cerebrospinal fluid. There are also few genetic factors like the presence of gene apolipoprotein that increases the risk of the disease. Such biomarkers cause memory interruption in the brain since neurons

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10.1016/j.procs.2020.01.071

are the basic communication units that transmit brain signals. Our human brain contains billions of nerve cells each composed of neurons that process and communicate information through electrical and chemical signals. They convey messages between different parts of the brain and from the brain to various parts of the body. When the pathological proteins accumulate in the brain and also the individual is with the genetic risk of APO protein, it leads to Alzheimer's disease that disrupts this communication between neurons, resulting in loss of normal functions of the brain and cell death. Hence the final stage of disease eventually causes neurons cell damage and death. The present research problem includes identifying biomarkers for disease, identifying patient's health and symptoms whether it falls under cognition disorder due to normal aging or MCI and dementia[8]. Mild cognitive impairment (MCI) is a condition distinguished by changes in brain functioning capabilities beyond the expected range of subject's age and education that are most normal and mild and they do not disrupt with its day to day activities. The multiclass classifier was constructed to predict the dataset of neuro-cognitive biomarkers. Now and then, the characterization by one explicit classifier does not give the ideal outcome because of specific variables, so there is a need to utilize multistage classifier for progressively right classification results measured by linear and logistic regression model[2].

It is important to classify the classes with abnormal cognitive decline related to normal aging in the early stage with the actual symptoms of the cognitive disorder for AD. Our proposed classifier showed the best results in such classification. In this proposed study, we develop a successful classification framework neurodegenerative disorder at the early onset of symptoms by utilizing the pre-processed input features, using various classifiers as an ensemble of classifiers, kernalizing the feature space for more accurate results. The features obtained through MRI scans used in this study were acquired from the Alzheimer's Disease Neuroimaging Institute (ADNI) and using Free Surfer an open-source software that pre-processes and analyses the neurobiological structure of brain through MRI images. In this manner, cerebrum auxiliary, volumetric measurements, and cortical thickness estimations were utilized as feature vectors for multiclass classification [3]. We analyzed the input feature set for missing data, inconsistent values and then to choose the best bio-markers which are early symptoms of normal aging or the actual Alzheimer's onset and constructed a multistage classifier.

The present work is organized as follows: Literature review, Proposed work, Dataset and methodology, MRI preprocessing, Classifier models such as KNN, SVM, and Gaussian Naive Bayes, Ensemble of classifiers, Experiments and results and Discussions.

2. Related work and literature review

The literature study was performed on Artificial Intelligence, Machine learning works implemented in providing solutions for medical applications. Many studies were performed on expert system solutions for Alzheimer's disease and Mild Cognitive Impairment conversions. The selection of features for such systems depends on choosing necessary machine learning algorithms, classifiers, neural network and deep learning algorithms significant in the prediction of AD[4]. Neuroimaging studies provide structural and functional changes occurring in the brain due to aging and also provides possible conversion rate of AD individuals from the cognitively impaired elderly community[7]. Several research studies utilized structural changes of the brain from MRI imaging techniques and functional changes from PET imaging and both are the biomarkers for AD detection at early onset of disease[5]. Many experiments has been made in MRI imaging feature extraction. The hippocampus volume data is utilized to monitor structural changes in brain that occur due to neuron degeneration. It acts as an important biomarker for disease identification that is caused by abnormal protein accumulation in neurons.[6]. Such proteins are beta amyloid and tau accumulations that block neurons and its functionality and finally cause cell death. The volume of brain is also the changing feature with disease progression due to change in the hippocampus and cortical brain regions. Changes in structure, volume and thickness are due to neuron degeneration, cell death and also due to genetic mutations that occur several years before the actual disease progression. Due to initial brain changes brain manages without showing any disease symptoms and continuous brain cell damage caused more behavioral abnormalities.

3. Proposed work

The proposed method is introduced with the purpose of early prediction of the disease and classification. It includes :(i) to find a learning model that efficiently differentiates AD from MCI, Pre-MCI, and NC (ii) to develop a novel classifier method and validate performance using regression. The MRI scans used in the proposed work are obtained from the Alzheimer's Disease

Neuroimaging Institute (ADNI), Laboratory of Neuro Imaging (LONI). Neuro Imaging data can also be obtained from OASIS by running XNAT scripts. Given in Table 1 is feature size of AD pathology stages of testing and training data. From the data set, feature scaling is performed with structural changes, volume and thickness of cortex to be compatible for input vectors for the proposed classifier hypothesis. Best biomarkers are selected for differentiating AD, MCI, Pre MCI, and normal controls. Based on input features we build a novel classifier algorithm using Naive Bayes, K-nearest neighbors and SVM, choosing kernel of which is dependent on the number of features and training set of elderly inhabitants. It can be improved by choosing other supervised models such as logistic regression when the number of features varies.

Table 1: Feature size of AD pathology stages of test and train.

Class	Number of Features	Test/Train
Alzheimer's disease (AD)	170	80:20
Mild Cognitive Impairment (MCI)	160	80:20
Normal Controls (NC)	130	80:20

3.1. Data and methodology

3.1.1. Dataset

The dataset utilized in this methodology is obtained from the ADNI database (Alzheimer's disease Neuro-Imaging Initiative). It provides a stand methodology for biomarker identification and supplies high data availability to researchers. It provides structural and functional changes of the brain in the shape of hippocampus regions of the brain and neurons. It also provides data about the deposition of Amyloid Beta protein and tau so that it can be utilized to check if it is abnormal or not. The cores of ADNI includes ADNI-1, ADNI-Go, ADNI-2. ADNI also provides pathological and genetic features of disease towards disease progression. These features lead ADNI as the large utilized database for data sharing by worldwide researchers. It provides features in three stages. The MRI data (Magnetic Resonance Imaging), PET (Positron Emission Tomography) and genetic arrangements about disease progression.

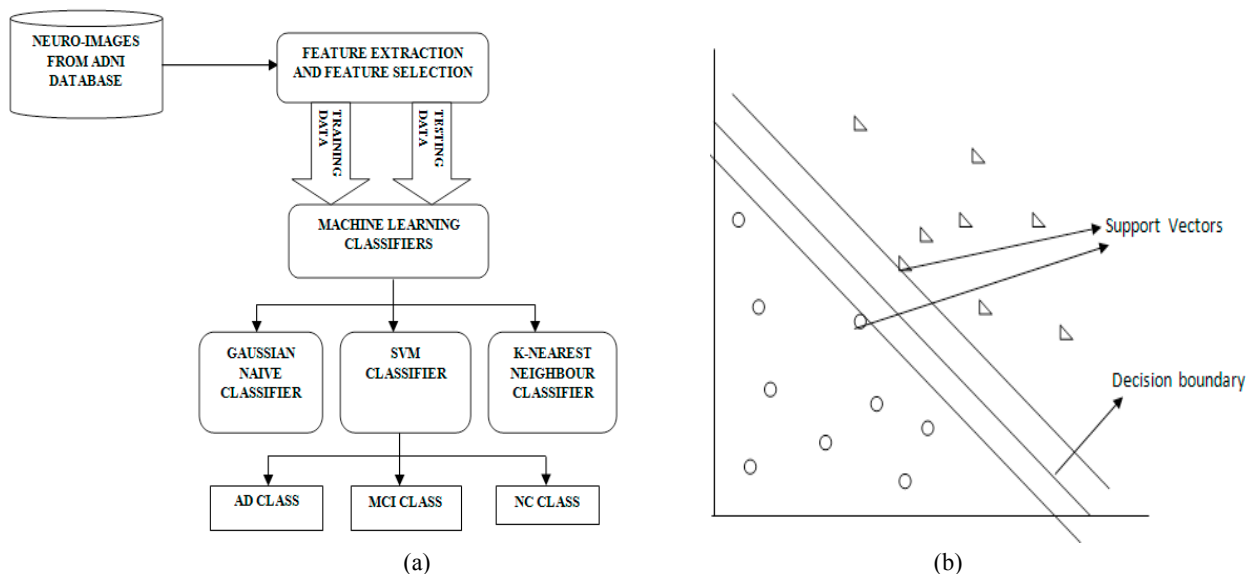


Fig. 1 (a) Ensemble classifier for Detecting AD; (b) Support Vector Machine classifier

3.1.2 MRI pre-processing

Using all the input features, training was performed initially. The features include three higher dimensional feature vectors including volume and thickness measurements of brain that specifies the structural features of brain and the aggregate of them; these may incorporate noisy(not helpful information in AD or MCI discovery) and excess data. Previous research and clinical encounters were considered using statistical examination, for creating results to choose a component rich information portrayal for AD or MCI Only. Excluding non-significant and excess features and lessening the dimensionality of the calculation results in more accurate performance. SVM, GP, KNN, and LR don't naturally work any input feature selection during training thus, for these strategies, we reperformed the preparation and hyper parameter streamlining process with two diminished sets of parameters using free surfer.. Fig.2 represents MRI preprocessing results.

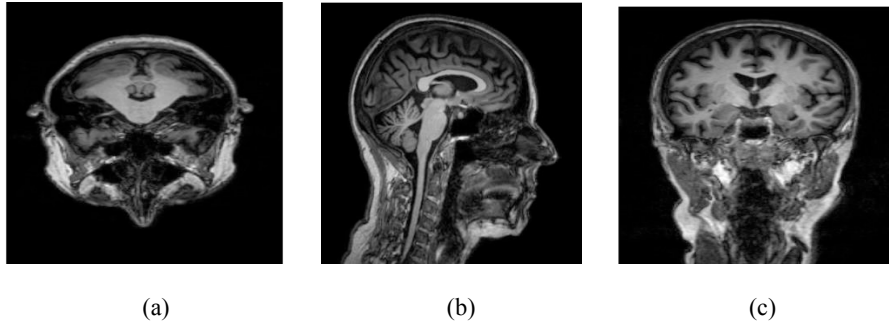


Fig.2.MRI Pre-processing-(a) Axial (b) Sagittal (c) Coronal

3.1.4. Ensemble of classifiers

In the present study, three classifier models are implemented and novel classifier is proposed by combining the classifiers in a single model that includes KNN, SVM, and Gaussian Naive Bayes classifier model as in fig.1.

Gaussian Naive Bayes classifier:

Bayesian learning is applied for learning the structure of data in the proposed problem. Bayesian techniques give a few auxiliary learning calculations. They give models of normal results and enable us to investigate causal connections, perform illustratively were c is the new case, e is a known example and i is every one of the N input features. To make the expectation, the k less distant models, additionally called its closest neighbors, is considered and class prediction of Alzheimer's disease or Mild cognitive impairment or Normal healthy controls are performed considering the quantity of closest neighbors. The proposed classification techniques are applied to data prediction of features between normal controls and AD groups. Apart from binary classification there can be more classification among features such as AD, Pre-mild MCI, MCI, cognitively normal aged individuals. This can be implemented by training the proposed ensemble of classifiers with hypothesis for each class of disease to predict probability and to make prediction on new disease class by choosing the class that has maximum value for hypothesis. Classifiers are implemented in supervised learning and can be utilized in classification of features that differ from normal aging. The fundamental advantage of Gaussian classifier is that it requires a little measure of training data which is important for characterization. For the particular list of features for each class a hypothesis is derived and mean, variance is determined where an ordinary appropriation is then parameterized as follows,

$$h_0(y) = \Theta_0 + \Theta_1 x = 1/2M \sum_{i=0}^m (h_0(x) - y)^2 \quad (1)$$

$$p(v, c) = \frac{1}{\sqrt{2\pi\sigma_{cl}^2}} * e^{-\frac{(v - \mu_{cl})^2}{2\sigma_{cl}^2}} \quad (2)$$

where, $\Theta_0 + \Theta_1$ are chosen so that $h_0(y)$ is close to the predicted class from our training examples, cl is the class to be labeled, v is learning rate, σ_{cl}^2 and μ_{cl} are variance and mean of the predicted class from hypothesis. This equation predicts how the

choice of parameters affects the accuracy in three different classification. Based on this Gaussian data, training is done as process of extracting model parameters. Once the model fitting, the trained model is validated.

K-Nearest Neighbors (kNN):

Simplest machine learning classifier KNN is applied for feature partitioning. In the kNN, initially the distances or the dissimilarity between a testing data of normal control with the training data and known data of all known control groups (i.e., those included in the training set whose output is whether AD, NC or MCI) is calculated. In this distance analysis, the Euclidean distance was used as distance metric, that is size of the training sample and requires optimization during the training phase. Here, K is the non-linear parameter which is instance based classifier for the algorithm that. In this method, the nearest neighbor is being tested and data point X_{train} , Y_{train} are calculated where as x_i , $i = 1, \dots, n$.

$$d_E(x, x_i) = \sum_{i=1}^n |x - x_i| \quad (3)$$

$$x \rightarrow d_E(x, x_i) < d_E(x, x_j), i \neq j \quad (4)$$

The closest k neighbors are resolved. Testing data indicates are characterized with respect to determined k closest neighbors. This classifier in our proposed study transforms real valued feature to an output between 0 and 1, that is evaluated as probability of input feature that belong to positive class given its input features $(x_0, x_1, \dots, x_n)$.

Support vector machine (SVM):

This supervised learning model is used for classification. In SVM, the normalization is done with input of real dataset of AD.SVM import support vector classifiers by using X_{test} and X_{train} that are advantageous in classifying the large number of input features obtained from MRI imaging and genetic information of aged inhabitants. Fig 1 represents the SVM graph that classifies features under multiple disease classes. By implementing SVM, choice of parameter c and similarity function are specified that depends on the number of input features and training examples .It groups the data by a hyper plane that is given as w^T and b of one dimensional to high-dimensional feature space. b is given as bias for the input vector x , and w is the weight of each feature. A good partition is the choosing the nearest training data points falling within the features of any of the three classes. The more the distance of the edge, the lower will be the generality error of the classifier. Assuming N training data samples $\{(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)\}$ are given, the classification problem can be given by the following minimization problem:

$$\min_{w,b} \left(\frac{(w^T * w)}{2} \right) \quad (5)$$

The ensemble of three classifiers

The front stage classifier characterizes the feature of any class among AD, MCI and NC, the following stage classifier isn't required and the classification procedure ends. If the classification procedure of the front stage is lacking, further stages are expected to get progressively exact classification results. Initially the Gaussian Naive Bayes Classifier classifies the probability of whether the feature belongs to one of the labels whose are the causes of Alzheimer's dementia or Alzheimer's MCI or normal aging or healthy control groups followed by SVM classifier and KNN classifier, classifies the features depending on the performance basis of 1st stage classifier and it chooses the type of kernel for parameter selection and model complexity[10].First classifier model predicts as an AD class that is more accurate with present literature studies. If the first classifier model labels the features as MCI and AD class with more probability, we input this result to second classifier (SVM and KNN classifier) and then analyze the classification results generated by this 2nd classifier and validated by linear and logistic regression models choosing various different features from the dataset. If the result of the first stage is uncertain, this object is transferred to a 2nd classifier (KNN classifier in this case). The original dataset divided into 5 parts of equal folds for last stage validation. The KNN classifier is trained as a multiclass classifier (for the order of all AD, MCI, and NC). This trains the model with multi-dimensional feature space for accurate classification. Accordingly, Gaussian Naive

Bayes Classifier is prepared as a parallel classifier (for recognition of AD, MCI, and NC) and the SVM classifier is likewise trained as a multiclass classifier (for prediction of AD, MCI or NC). Regularization for SVM is checked with the C parameter that determines the strength of regularization. As the value of C is larger, less the regularization and feature fits the training data.

Table: 2: Given rows are predicted groups and columns are true classification.

a)Classification in terms of Volume of brain and Thickness of brain

Class	Volume of brain			Thickness of brain		
	NC	AD	MCI	NC	AD	MCI
Alzheimer's disease(AD)	7	22	5	4	27	3
Normal Controls(NC)	21	2	3	21	2	3
Mild Cognitive Impairment(MCI)	4	3	25	3	2	27
b)Total						
Class	NC	AD	MCI			
Alzheimer's disease(AD)	2	30	2			
Normal Controls(NC)	19	3	4			
Mild Cognitive Impairment(MCI)	3	2	27			

4. Experiments and results

The present study used system resource consisting of a core i3 processor of 2.3 GHz speed, 8 GB RAM and the Windows operating system. Classifiers are implemented in Jupyter notebook, linear and logistic regression in GNU Octave. Libraries included are SciPy, NumPy, Pandas, and Matplotlib. The data sets were obtained from ADNI-LONI that includes 136 AD features, 128 MCI features and 104 NC features for training model, and 34 AD features, 32 MCI features and 26 healthy controls for testing the performance of prediction by the proposed model as given in Table.2. The names of the volume and thickness feature selected by feature extraction.

```

Program paused. Press enter to continue.
Cost at initial theta (zeros): 0.693147
Expected cost (approx): 0.693
Gradient at initial theta (zeros):
-0.100000
-12.009217
-11.262842
Expected gradients (approx):
-0.1000
-12.0092
-11.2628
Cost at test theta: 0.218330
Expected cost (approx): 0.218
Gradient at test theta:
0.042903
2.566234
2.646797
Expected gradients (approx):
0.043
2.566
2.647
Program paused. Press enter to continue.
Cost at theta found by fminunc: 0.203498
Expected cost (approx): 0.203
theta:
-25.161272
0.206233
0.201470
Expected theta (approx):
-25.161
0.206
0.201
Program paused. Press enter to continue.
For given features with AD symptoms and normal aging, we predict probability of 0
.776289
Expected value: 0.775 +/- 0.002
Train Accuracy: 89.000000
Expected accuracy (approx): 89.0
octave:9> _

```

Fig.3.Accuracy of ML classifier

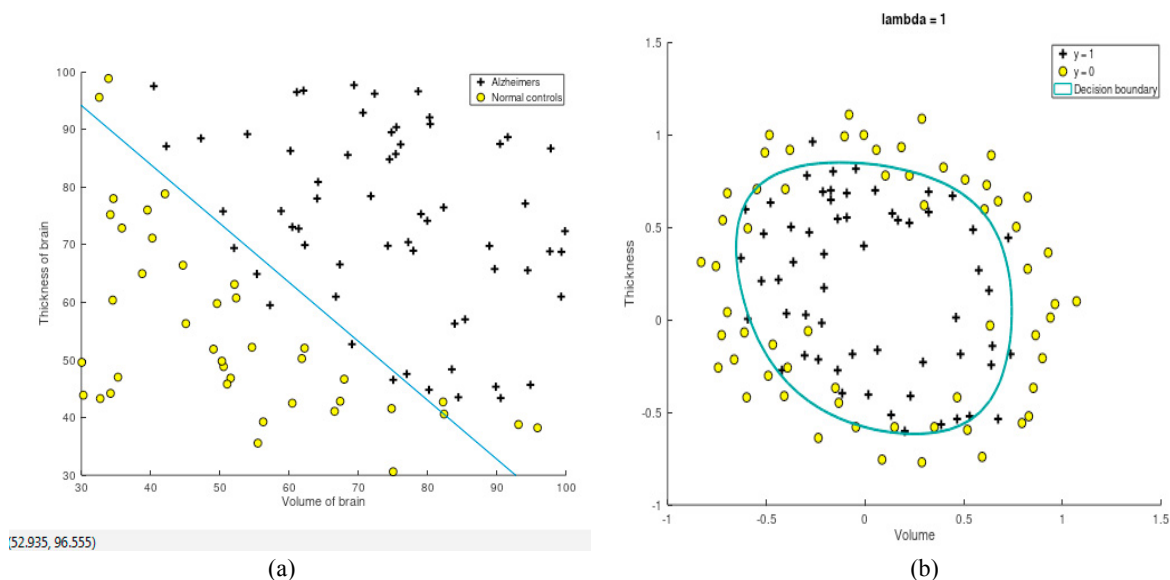


Fig.4.Validation test results (a) Linear regression (b) Logistic regression

With these given features the training with labels Y_{train} and so that given new instance X_{test} are classified. The accuracy and validation results are given in Fig.3 and Fig.4. The most similar memory feature to the test that is in X_{train} is classified into a particular class. This is achieved by specifying the nearest neighbor parameter, distance metric, weighing function on nearest points, aggregating among three classes based on the nearest points.

5. Discussions

Existing machine learning and deep learning algorithms derive only limited predictive accuracy since they employ complex features including socio demographic characters, MMSE score and genetic combinations. They derive results based on expensive machine learning algorithms that use these features and hence classification becomes complex with when decision boundaries are obtained. It is addressed in the proposed study by the choice of powerful and better features such as MRI scan data of neuron degeneration, ensemble classification model for differentiation between normal aging and AD. Structural changes of brain including volume and thickness are the first brain changes occurring for AD, hence such features makes our proposed study to yield accurate result and prediction of disease at earlier stage. The prediction score are given in Table.3. It has proved high accuracy values utilizing the sci-kit-learn library. In data set, features are analyzed and choice utilized in this examination was performed to mirror the structural changes in the brain, which is identified with the clinical advancement of AD. Since the features are simple, it fits best for the proposed learning hypothesis in classifier algorithm and kernel selection for SVM.

Table.3: Prediction Score

Class	AD	MCI	NC
Thickness	0.88	0.88	0.86
Volume	0.83	0.85	0.80
Total	0.88	0.89	0.88

6. Conclusion and Future Work

The multiclass classifier utilized in this proposed study resulted in the best results for AD class differentiation as compared to present individual machine learning frameworks. Additionally, the feature selection, Kernel and parameter selection, following the proposed strategy for AD classification, delivered great outcomes. Significant enhancements were seen in recovery speed and accuracy during the execution. In any case, in spite of these potential improvements, there are a few issues for segregating MCI and NC classes versus AD. The false alarm rate for the AD class was less when compared to other different classes - NC and MCI. This implies there is in need of advanced features in disease predictions like genetic data, glucose level of cerebrospinal fluid that are expected to combine with MRI biomarkers for better performance in distinguishing the MCI, AD and NC classes [9]. Also, the choice of initial classifier model is significant in the design of the proposed ensemble classifier. The multiclass classifier was constructed as one vs all classifier to predict neuro-cognitive biomarkers. By the characterization, by one explicit classifier model that does not give the ideal outcome because of specific variables, so the collection of many classifier models as multistage classifier model utilized for progressively best result in distinguishing different stages of pathology. This can also be further improved by improving parameters such as choosing conjugate gradient, BFGS, L-BFGS that are more optimized procedures than presently worked gradient descent in all classifier models. It is important to differentiate the classes with abnormal cognitive decline related to normal aging in an early stage with the actual symptoms of the cognitive disorder for AD. Our proposed classifier showed the best result in such classification.

References

- [1] Brooks LG, Loewenstein DA (2010) Assessing the progression of mild cognitive impairment to Alzheimer's disease: Current trends and future directions. *Alzheimers Res Ther*, 2, 28.
- [2] Cooper C, Sommerlad A, Lyketsos CG, Livingston G (2015) Modifiable predictors of dementia in mild cognitive impairment: A systematic review and meta-analysis. *Am J Psychiatry* 172, 323-334.
- [3] Kang JH, Korecka M, Toledo JB, Trojanowski JQ, Shaw LM (2013) Clinical utility and analytical challenges in measurement of cerebrospinal fluid amyloid-beta(1-42) and tau proteins as Alzheimer disease biomarkers. *Clin Chem* 59, 903-916.
- [4] Kruthika et al. (2019) Multistage classifier-based approach for Alzheimer's disease prediction and retrieval, *Informatics in Medicine Unlocked*, 14:34-42.
- [5] Loewenstein DA, Curiel RE, Duara R, Buschke H (2017) Novel cognitive paradigms for the detection of memory impairment in preclinical Alzheimer's disease. *Assessment*, 1073191117691608.
- [6] Petersen RC, Parisi JE, Dickson DW, Johnson KA, Knopman DS, Boeve BF, Jicha GA, Ivnik RJ, Smith GE, Tangalos EG, Braak H, Kokmen E (2006) Neuropathologic features of amnesic mild cognitive impairment.
- [7] Plant C, Teipel SJ, Oswald A, Bohm C, Meindl T, Mourao-Miranda J, Bokde AW, Hampel H, Ewers M (2010) Automated detection of brain atrophy patterns based on MRI for the prediction of Alzheimer's disease. *Neuroimage* 50, 162-174.
- [8] Sperling R, Johnson K (2013) Biomarkers of Alzheimer disease: Current and future applications to diagnostic criteria. *Continuum (Minneapolis)* 19, 325-338.
- [9] Van Rossum IA, Vos S, Handels R, Visser PJ (2010) Biomarkers as predictors for conversion from mild cognitive impairment to Alzheimer-type dementia: Implications for trial design. *J Alzheimers Dis* 20, 881-891.
- [10] Van Cauwenberghe C, Van Broeckhoven C, Sleegers K (2016) The genetic landscape of Alzheimer disease: Clinical implications and perspectives. *Genet Med* 18, 421-430.